CHE210: *DISCUSSION SESSION 3* **NAME:**___________________________________

FEBRUARY 2, 2020

Evaluate the streams:

1. 75mL CCl_4 (SG=1.59)

   ⟶ Calculate n (mol CCl_4)

2. m = 100g/h

   ⟶ Calculate m_{C2H4}

   x_{CH4} = 0.3 (CH_4 mol/mol)

   x_{C2H4} = 0.4 (C_2H_4 mol/mol)

   x_{C2H6} = 0.3 (C_2H_6 mol/mol)

3. n = 250 kmol/h

   ⟶ Calculate n_{H20} [in terms of x]

   ω_{H20} = x [kg H_2O/kg]

   ω_{H2} = (1 - x) [kg H_2/kg]

P2.43 *Mixtures of hydrocarbon gases and air will burn only if the gas:air ratio is within the flammability limit. For example, propane gas-air will not ignite, even if exposed to flame, if the propane composition of the mixture is greater than 11.4 mol%. Calculate the pounds per hour of propane (C_3H_8) to mix with 100L/hr of air (at 1 atm pressure and 32°F) to just exceed the flammability limit and avoid ignition of the propane-air mixture. Air can be assumed to be 79 mol% N_2 and 21 mol% O_2. The density of air or propane gas can be calculated from the ideal gas law.*